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CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 89-162

WASTE DISCHARGE REQUIREMENTS
FOR
FLYING J. INC.
FLYING 'J' TRAVEL PLAZA
KERN COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Flying J. Inc. submitted a Report of Waste Discharge dated 21 December 1988 and technical report dated 31 March 1989 for a discharge of domestic sewage from Flying 'J' Travel Plaza, a proposed truck stop. The truck stop will include a motel, service station, restaurant, and store near State Highway 99 and Merced Avenue, as shown in Attachment A. The site is in the southeast quarter of Section 32, T27S, R26E, MDB&M, about 11 miles north of Bakersfield and 1-1/4 miles east of Lerdo Canal.
2. Treatment will be provided by an extended aeration package plant. Five percolation/evaporation ponds having a total surface area of 0.253 acres will be used for disposal. An aeration basin is available for emergencies, such as power outages.
3. The motel, service station, restaurant, and store will generate an average daily dry weather discharge flow of 0.043 mgd. The 30-day average dry weather design flow of the treatment and disposal facilities is 0.045 mgd. The maximum daily flow capacity of the treatment and disposal facilities is 0.060 mgd.
4. Soils beneath the site consist of silty sands, with small amounts of gravel, to depths of 15 feet.
5. Unconfined ground water occurs at a depth of about 275 feet below ground surface and is of poor quality, with a specific electrical conductance (EC) of about 1340 umhos/cm and a nitrate concentration of about 124 mg/l. The State Recommended Secondary Maximum Contaminant Level (MCL) for EC is 900 umhos/cm and the State Recommended Primary MCL for nitrate is 45 mg/l. Confined (deeper) ground water is of a higher quality, with an EC of about 250 umhos/cm and a nitrate concentration of about 40 mg/l.
6. The beneficial uses of ground water are municipal and domestic supply and agricultural supply.
7. Poso Creek is about four miles north. The beneficial uses of Poso Creek include agricultural supply, recreation, fresh water habitat, preservation of rare and endangered species, ground water recharge and fresh water replenishment.

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8. The Board, on 21 August 1975, adopted the Water Quality Control Plan for the Tulare Lake Basin (5D). These requirements are consistent with that plan.
9. The County of Kern approved a Negative Declaration for the project in accordance with the California Environmental Quality Act (Public Resources Code, Section 21000, et seq.). The Board has reviewed the negative declaration and concurs there will be no significant impacts on water quality.
10. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for the discharge and provided them with an opportunity to submit views and recommendations.
11. The Board in a public meeting heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Flying J. Inc., in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following at Flying 'J' Travel Plaza:

A. Discharge Prohibitions:

1. The discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. The by-pass or overflow of untreated or partially treated waste is prohibited.
3. Acceptance of septage, and wastes other than domestic wastewater generated by the truck stop is prohibited.

B. Discharge Specifications:

1. Neither the treatment nor the discharge shall cause a pollution or nuisance as defined by the California Water Code, Section 13050.
2. The discharge shall not cause degradation of any water supply.
3. The discharge shall remain within the designated disposal area (ponds) at all times.
4. The monthly average daily discharge flow shall not exceed 45,000 gallons.

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5. The discharge from the package treatment plant to the ponds shall comply with the following limitations:

<u>Constituent</u>	<u>Monthly Mean</u>	<u>Maximum</u>
BOD ₅ , mg/l	30	90
Settleable Solids, ml/l	0.2	1.0

6. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner consistent with Subchapter 15 of Title 23, California Code of Regulations, as approved by the Executive Officer.
7. The dissolved oxygen content of the upper 0.5 feet of the percolation/evaporation ponds shall not be less than 1.0 mg/l for 16 hours in any 24-hour period.
8. The disposal ponds shall be operated with a minimum freeboard of 2 feet at all times.
9. The maximum specific electrical conductance of the discharge shall not exceed the weighted average of the source water plus 500 micromhos/cm.
10. Floating scum, grease, and other materials on pond surfaces shall be removed regularly.
11. Disposal of wastewater shall not occur within 100 feet of any domestic well or within 50 feet of any irrigation well.
12. The evaporation/percolation ponds shall have sufficient capacity to accommodate wastewater and design seasonal precipitation. Design seasonal precipitation shall be based on total annual precipitation using a return period of 25 years, distributed monthly in accordance with historical rainfall patterns.
13. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment area.
14. Public contact with wastewater shall be precluded through such means as fences and signs.

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C. Provisions:

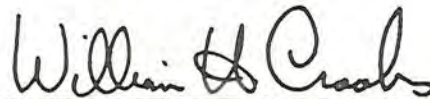
1. The Discharger shall submit technical reports as directed by the Executive Officer.
2. The Discharger shall comply with Monitoring and Reporting Program No. 89-162, which is attached hereto and part of this Order.
3. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 September 1985, which are attached hereto and part of this Order.
4. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.
5. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.
6. Physical facilities shall be designed and constructed according to accepted engineering practice and shall be capable of full compliance with this Order when properly operated and maintained. Proper operation and maintenance shall be described in an Operation and Maintenance Manual.
7. Use of the emergency aeration basin shall be limited to emergency situations only.
8. The Discharger shall submit the technical reports described in Standard Provision B.3, and a copy of the Operation and Maintenance Manual for the facility, by 15 November 1989.
9. A copy of these waste discharge requirements and the Operation Maintenance Manual shall be maintained at the facility and be available at all times to operating personnel.
10. The Discharger must comply with all conditions of the waste discharge requirements. Violations may result in enforcement actions, including Regional Board orders or court orders requiring corrective action or imposing civil monetary liability, or in modification or revocation of these waste discharge requirements by the Regional Board.

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11. The Board will review this Order periodically and revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 22 September 1989.



WILLIAM H. CROOKS, Executive Officer

LC:cjs:8/25/89

Attachments

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 89-162

FOR
FLYING J. INC.
FLYING 'J' TRAVEL PLAZA
KERN COUNTY

EFFLUENT MONITORING

Excepting as noted below, samples shall be collected from the discharge from the package plant. The time of collection of the grab sample shall be recorded. The following shall constitute the effluent monitoring program:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Total Daily Flow ^{1/}	mgd	Continuous	Daily
Dissolved Oxygen ^{2/}	mg/l	Grab	Daily
Specific Electrical Conductance	umhos/cm	Grab	Monthly ^{3/}
BOD ^{4/}	mg/l	Grab	Weekly
Settleable Solids	ml/l	Grab	Weekly

- 1/ Total daily flow may be measured in either the influent or effluent waste streams.
- 2/ To be collected opposite the inlet of each pond at a depth of about 0.5 feet between the hours of 0800 and 0900.
- 3/ Effluent to be sampled monthly. Source water to be sampled in January of each year. If the source water is from more than one well, the EC shall be a weighted average.
- 4/ Five-day, 20°C, biochemical oxygen demand (BOD).

FREEBOARD MONITORING

The freeboard shall be monitored on all evaporation/percolation ponds in use to the nearest one-half foot. The following shall constitute the freeboard monitoring program.

<u>Constituent</u>	<u>Unit</u>	<u>Type of Sample</u>	<u>Frequency</u>
Freeboard	feet	Observation	Weekly

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Permanent markers shall be placed in each pond with calibrations, to the nearest one-half foot, indicating the baseline freeboard of 2.0 feet and available operational freeboard.

REPORTING

Monitoring reports shall be submitted to the Regional Board by the 15th day of the month following the month of sampling. The reports will include date, time, duration, cause, and remedy on any emergency that necessitates use of the emergency aeration basin unless the Discharger has already reported pursuant to Standard Provision B.2.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in a manner that illustrates clearly whether the discharge complies with waste discharge requirements.

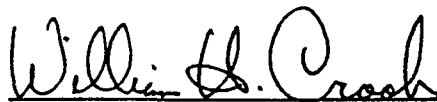
The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board.

The Discharger may also be requested to submit an annual report to the Board with both tabular and graphical summaries of the monitoring data obtained during the previous year. (Any such request will be made in writing.) The report shall discuss the compliance record. If violations have occurred, the report shall also discuss the corrective actions taken and planned to bring the discharge into full compliance with waste discharge requirements.

All reports submitted in response to this Order shall comply with the signatory requirements of Standard Provision D.6.

The Discharger shall implement the above monitoring program effective 1 October 1989 when discharge begins, whichever is later.

Ordered by



WILLIAM H. CROOKS, Executive Officer

22 September 1989
(Date)

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INFORMATION SHEET

FLYING J. INC.
FLYING 'J' TRAVEL PLAZA
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Flying J. Inc. proposes to discharge domestic wastewater from a 42-acre motel, service station, restaurant, and store. The site, to be called Flying 'J' Travel Plaza, is on the east side of State Highway 99 and the north side of Merced Avenue, about 11 miles north of Bakersfield, in a portion of Section 32, T27S, R26E, MDB&M.

The improvements are designed to serve 335 people daily. The commercial operation will generate an average daily dry weather flow of 0.043 mgd. The 30-day average dry weather design flow of the treatment and disposal facilities is 0.045 mgd. The maximum daily flow capacity of the treatment facility is 0.060 mgd.

Land to the north, south and east is in agricultural preserve. To the west is State Highway 99 and more farmland.

The treatment facilities include a comminutor, bar screen, emergency aeration pond, and extended aeration package treatment plant. Effluent limits are based on rapid percolation, the close proximity of occupied dwellings and eating establishments, and Basin Plan guidance. The design criteria of the plant are more stringent than the effluent limits, so the system will be able to comply with effluent limits, if properly operated. Five percolation/evaporation ponds, covering a total of 0.253 acres and from 3.5 to 4.0 feet deep, will be used for disposal. The discharger has indicated that wastewater reclamation is not feasible because of the small volume of the discharge.

The 1987 Kern County Water Agency Water Supply Report indicates that unconfined ground water in the area is at a depth of about 275 feet below ground surface (b.g.s.) and flows in a southwesterly direction. The Kern County Health Department Laboratory analyzed unconfined ground water beneath the site and reported a specific conductance (EC) of 1340 umhos/cm and a nitrate concentration of 124 mg/l. The State Recommended Secondary Maximum Contaminant Level (MCL) for EC is 900 umhos/cm. The State Recommended Primary MCL for Nitrate is 45 mg/l. Confined ground water is of a higher quality (EC about 250 umhos/cm, nitrates about 40 mg/l) at more than 450 feet b.g.s.

Percolation tests indicate that the average annual soil percolation rate in the disposal area is 1.4 minutes per inch. Soils beneath the proposed disposal site are described as silty sands and clean sands, with small amounts of gravel, to depths of about 15 feet b.g.s.

One domestic well will be constructed 735 feet west of the Travel Plaza.

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The 1987 Water Supply Report indicates that the total annual pan evaporation was 74.42 inches and the annual precipitation was 6.8 inches, at the Bakersfield station. The long-term average precipitation (from 1963 to 1987) was 5.5 inches at the station.

The County of Kern approved a Negative Declaration for the project in accordance with the California Environmental Quality Act (Public Resources Code Section 21000 et seq.). The Board has reviewed the negative declaration and concurs there will be no significant impacts on water quality.

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